statistics statistical methods data analysis probability Research Report

Research Topic: statistics statistical methods data analysis probability

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Total Papers: 20

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Search Summary

Parameter	Value
Search Query	statistics statistical methods data analysis probability
Total Papers Found	20
Data Sources Used	arXiv, CrossRef
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Core Papers List

Paper #1

Title: The Statistical Analysis of fMRI Data

Authors: Martin A. Lindquist

Year: 2009

Abstract: In recent years there has been explosive growth in the number of neuroimaging studies performed using functional Magnetic Resonance Imaging (fMRI). The field that has grown around the acquisition and analysis of fMRI data is intrinsically interdisciplinary in nature and involves contributions from researchers in neuroscience, psychology, physics and statistics, among others. A standard fMRI study gives rise to massive amounts of noisy data with a complicated spatio-temporal correlation structure...

URL: http://arxiv.org/abs/0906.3662v1

Source: arXiv

Paper #2

Title: Statistical Inference: The Big Picture

Authors: Robert E. Kass

Year: 2011

Abstract: Statistics has moved beyond the frequentist-Bayesian controversies of the past. Where does this leave our ability to interpret results? I suggest that a philosophy compatible with statistical practice, labeled here statistical pragmatism, serves as a foundation for inference. Statistical pragmatism is inclusive and emphasizes the assumptions that connect statistical models with observed data. I argue that introductory courses often mischaracterize the process of statistical inference and I propo...

URL: http://arxiv.org/abs/1106.2895v2

Source: arXiv

Paper #3

Title: Statistical Methods in Topological Data Analysis for Complex, High-Dimensional Data

Authors: Patrick S. Medina, R. W. Doerge

Year: 2016

Abstract: The utilization of statistical methods an their applications within the new field of study known as Topological Data Analysis has has tremendous potential for broadening our exploration and understanding of complex, high-dimensional data spaces. This paper provides an introductory overview of the mathematical underpinnings of Topological Data Analysis, the workflow to convert samples of data to topological summary statistics, and some of the statistical methods developed for performing inference...

URL: http://arxiv.org/abs/1607.05150v1

Source: arXiv

Paper #4

Title: Statistics, Causality and Bell's Theorem

Authors: Richard D. Gill

Year: 2012

Abstract: Bell's [Physics 1 (1964) 195-200] theorem is popularly supposed to establish the nonlocality of quantum physics. Violation of Bell's inequality in experiments such as that of Aspect, Dalibard and Roger [Phys. Rev. Lett. 49 (1982) 1804-1807] provides empirical proof of nonlocality in the real world. This paper reviews recent work on Bell's theorem, linking it to issues in causality as understood by statisticians. The paper starts with a proof of a strong, finite sample, version of Bell's inequali...

URL: http://arxiv.org/abs/1207.5103v6

Source: arXiv

Paper #5

Title: A statistical framework for the analysis of microarray probe-level data

Authors: Zhijin Wu, Rafael A. Irizarry

Year: 2007

Abstract: In microarray technology, a number of critical steps are required to convert the raw measurements into the data relied upon by biologists and clinicians. These data manipulations, referred to as preprocessing, influence the quality of the ultimate measurements and studies that rely upon them. Standard operating procedure for microarray researchers is to use preprocessed data as the starting point for the statistical analyses that produce reported results. This has prevented many researchers from...

URL: http://arxiv.org/abs/0712.2115v1

Source: arXiv

Paper #6

Title: Elliptical Insights: Understanding Statistical Methods through Elliptical Geometry

Authors: Michael Friendly, Georges Monette, John Fox

Year: 2013

Abstract: Visual insights into a wide variety of statistical methods, for both didactic and data analytic purposes, can often be achieved through geometric diagrams and geometrically based statistical graphs. This paper extols and illustrates the virtues of the ellipse and her higher-dimensional cousins for both these purposes in a variety of contexts, including linear models, multivariate linear models and mixed-effect models. We emphasize the strong relationships among statistical methods, matrix-algebr...

URL: http://arxiv.org/abs/1302.4881v1

Source: arXiv

Paper #7

Title: Special section: Statistical methods for next-generation gene sequencing data

Authors: Karen Kafadar

Year: 2012

Abstract: This issue includes six articles that develop and apply statistical methods for the analysis of gene sequencing data of different types. The methods are tailored to the different data types and, in each case, lead to biological insights not readily identified without the use of statistical methods. A common feature in all articles is the development of methods for analyzing

simultaneously data of different types (e.g., genotype, phenotype, pedigree, etc.); that is, using data of one type to info...

URL: http://arxiv.org/abs/1206.6617v1

Source: arXiv

Paper #8

Title: Byzantine-Resilient SGD in High Dimensions on Heterogeneous Data

Authors: Deepesh Data, Suhas Diggavi

Year: 2020

Abstract: We study distributed stochastic gradient descent (SGD) in the master-worker architecture under Byzantine attacks. We consider the heterogeneous data model, where different workers may have different local datasets, and we do not make any probabilistic assumptions on data generation. At the core of our algorithm, we use the polynomial-time outlier-filtering procedure for robust mean estimation proposed by Steinhardt et al. (ITCS 2018) to filter-out corrupt gradients. In order to be able to apply ...

URL: http://arxiv.org/abs/2005.07866v1

Source: arXiv

Paper #9

Title: Statistical Modeling of RNA-Seq Data

Authors: Julia Salzman, Hui Jiang, Wing Hung Wong

Year: 2011

Abstract: Recently, ultra high-throughput sequencing of RNA (RNA-Seq) has been developed as an approach for analysis of gene expression. By obtaining tens or even hundreds of millions of reads of transcribed sequences, an RNA-Seq experiment can offer a comprehensive survey of the population of genes (transcripts) in any sample of interest. This paper introduces a statistical model for estimating isoform abundance from RNA-Seq data and is flexible enough to accommodate both single end and paired end RNA-Se...

URL: http://arxiv.org/abs/1106.3211v1

Source: arXiv

Paper #10

Title: Analyzing complex functional brain networks: fusing statistics and network science to understand the brain

Authors: Sean L. Simpson, F. DuBois Bowman, Paul J. Laurienti

Year: 2013

Abstract: Complex functional brain network analyses have exploded over the last eight years, gaining traction due to their profound clinical implications. The application of network science (an interdisciplinary offshoot of graph theory) has facilitated these analyses and enabled examining the brain as an integrated system that produces complex behaviors. While the field of statistics has been integral in advancing activation analyses and some connectivity analyses in functional neuroimaging research, it ...

URL: http://arxiv.org/abs/1302.5721v3

Source: arXiv

Paper #11

Title: Statistical Methods for Astronomy

Authors: Eric D. Feigelson, G. Jogesh Babu

Year: 2012

Abstract: This review outlines concepts of mathematical statistics, elements of probability theory, hypothesis tests and point estimation for use in the analysis of modern astronomical data. Least squares, maximum likelihood, and Bayesian approaches to statistical inference are treated. Resampling methods, particularly the bootstrap, provide valuable procedures when distributions functions of statistics are not known. Several approaches to model selection and good- ness of fit are considered. Applied stat...

URL: http://arxiv.org/abs/1205.2064v1

Source: arXiv

Paper #12

Title: The Golden Age of Statistical Graphics

Authors: Michael Friendly

Year: 2009

Abstract: Statistical graphics and data visualization have long histories, but their modern forms began only in the early 1800s. Between roughly 1850 and 1900 (±10), an explosive growth occurred in both the general use of graphic methods and the range of topics to which they were applied. Innovations were prodigious and some of the most exquisite graphics ever produced appeared, resulting in what may be called the ``Golden Age of Statistical Graphics.'' In this article I trace the origins of this peri...

URL: http://arxiv.org/abs/0906.3979v1

Source: arXiv

Paper #13

Title: Calibrated Bayes, for Statistics in General, and Missing Data in Particular

Authors: Roderick Little

Year: 2011

Abstract: It is argued that the Calibrated Bayesian (CB) approach to statistical inference capitalizes on the strength of Bayesian and frequentist approaches to statistical inference. In the CB approach, inferences under a particular model are Bayesian, but frequentist methods are useful for model development and model checking. In this article the CB approach is outlined. Bayesian methods for missing data are then reviewed from a CB perspective. The basic theory of the Bayesian approach, and the closely ...

URL: http://arxiv.org/abs/1108.1917v1

Source: arXiv

Paper #14

Title: Resolving conflicts between statistical methods by probability combination: Application to empirical Bayes analyses of genomic data

Authors: David R. Bickel

Year: 2011

Abstract: In the typical analysis of a data set, a single method is selected for statistical reporting even when equally applicable methods yield very different results. Examples of equally applicable methods can correspond to those of different ancillary statistics in frequentist inference and of different prior distributions in Bayesian inference. More broadly, choices are made between parametric and nonparametric methods and between frequentist and Bayesian methods. Rather than choosing a single meth...

URL: http://arxiv.org/abs/1111.6174v1

Source: arXiv

Paper #15

Title: Advanced Data Analysis of Spontaneous Biophoton Emission: A Multi-Method Approach

Authors: M. Benfatto, L. De Paolis, L. Tonello, P. Grigolini

Year: 2025

Abstract: Ultra-weak photon emission (UPE) from living systems is widely hypothesized to reflect un-derlying self-organization and long-range coordination in biological dynamics. However, distin-guishing biologically driven correlations from trivial stochastic or instrumental effects requires a robust, multi-method framework. In this work, we establish and benchmark a comprehensive anal-ysis pipeline for photon-count time series, combining Distribution Entropy Analysis, Rényi entro-py, Detrended Fluctuati...

URL: http://arxiv.org/abs/2511.11080v1

Source: arXiv

Paper #16

Title: Data Encoding for Byzantine-Resilient Distributed Optimization

Authors: Deepesh Data, Linqi Song, Suhas Diggavi

Year: 2019

Abstract: We study distributed optimization in the presence of Byzantine adversaries, where both data and computation are distributed among m worker machines, t of which may be corrupt. The compromised nodes may collaboratively and arbitrarily deviate from their pre-specified programs, and a designated (master) node iteratively computes the model/parameter vector for generalized linear models. In this work, we primarily focus on two iterative algorithms: Proximal Gradient Descent (PGD) and Coordinate ...

URL: http://arxiv.org/abs/1907.02664v2

Source: arXiv

Paper #17

Title: Pao-Lu Hsu (Xu, Bao-lu): The Grandparent of Probability and Statistics in China

Authors: Dayue Chen, Ingram Olkin

Year: 2012

Abstract: The years 1910-1911 are auspicious years in Chinese mathematics with the births of Pao-Lu Hsu, Luo-Keng Hua and Shiing-Shen Chern. These three began the development of modern mathematics in China: Hsu in probability and statistics, Hua in number theory, and Chern in differential geometry. We here review some facts about the life of P.-L. Hsu which have been uncovered recently, and then discuss some of his contributions. We have drawn heavily on three papers in the 1979 Annals of Statistics (volu...

URL: http://arxiv.org/abs/1210.1031v1

Source: arXiv

Paper #18

Title: Statistical Modeling of Spatial Extremes

Authors: A. C. Davison, S. A. Padoan, M. Ribatet

Year: 2012

Abstract: The areal modeling of the extremes of a natural process such as rainfall or temperature is important in environmental statistics; for example, understanding extreme areal rainfall is crucial in flood protection. This article reviews recent progress in the statistical

modeling of spatial extremes, starting with sketches of the necessary elements of extreme value statistics and geostatistics. The main types of statistical models thus far proposed, based on latent variables, on copulas and on spati...

URL: http://arxiv.org/abs/1208.3378v1

Source: arXiv

Paper #19

Title: Dynamic Modeling and Statistical Analysis of Event Times

Authors: Edsel A. Peña

Year: 2007

Abstract: This review article provides an overview of recent work in the modeling and analysis of recurrent events arising in engineering, reliability, public health, biomedicine and other areas. Recurrent event modeling possesses unique facets making it different and more difficult to handle than single event settings. For instance, the impact of an increasing number of event occurrences needs to be taken into account, the effects of covariates should be considered, potential association among the intere...

URL: http://arxiv.org/abs/0708.0343v1

Source: arXiv

Paper #20

Title: Statistical Analysis in Genetic Studies of Mental Illnesses

Authors: Heping Zhang

Year: 2011

Abstract: Identifying the risk factors for mental illnesses is of significant public health importance. Diagnosis, stigma associated with mental illnesses, comorbidity, and complex etiologies, among others, make it very challenging to study mental disorders. Genetic studies of mental illnesses date back at least a century ago, beginning with descriptive studies based on Mendelian laws of inheritance. A variety of study designs including twin studies, family studies, linkage analysis, and more recently, ge...

URL: http://arxiv.org/abs/1108.3168v1

Research Trend Analysis

Publication Year Distribution

Year	Papers
2025	1
2020	1
2019	1
2016	1
2013	2
2012	5
2011	5
2009	2
2007	2

Data Source Distribution

Source	Papers
arXiv	20

Research Insights

• Total of 20 papers found matching the search criteria.

• Most recent publications from 2025.

• Primary data source: arXiv (20 papers).

References

[1] Martin A. Lindquist. The Statistical Analysis of fMRI Data. 2009.

[2] Robert E. Kass. Statistical Inference: The Big Picture. 2011.

[3] Patrick S. Medina, R. W. Doerge. Statistical Methods in Topological Data Analysis for Complex, High-Dimensional Data. 2016.

[4] Richard D. Gill. Statistics, Causality and Bell's Theorem. 2012.

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[6] Michael Friendly, Georges Monette, John Fox. Elliptical Insights: Understanding Statistical Methods through Elliptical Geometry. 2013.

[7] Karen Kafadar. Special section: Statistical methods for next-generation gene sequencing data. 2012.

[8] Deepesh Data, Suhas Diggavi. Byzantine-Resilient SGD in High Dimensions on Heterogeneous Data. 2020.

[9] Julia Salzman, Hui Jiang, Wing Hung Wong. Statistical Modeling of RNA-Seq Data. 2011.

[10] Sean L. Simpson, F. DuBois Bowman, Paul J. Laurienti. Analyzing complex functional brain networks: fusing statistics and network science to understand the brain. 2013.

[11] Eric D. Feigelson, G. Jogesh Babu. Statistical Methods for Astronomy. 2012.

[12] Michael Friendly. The Golden Age of Statistical Graphics. 2009.

[13] Roderick Little. Calibrated Bayes, for Statistics in General, and Missing Data in Particular. 2011.

[14] David R. Bickel. Resolving conflicts between statistical methods by probability combination: Application to empirical Bayes analyses of genomic data. 2011.

[15] M. Benfatto, L. De Paolis, L. Tonello, et al.. Advanced Data Analysis of Spontaneous Biophoton Emission: A Multi-Method Approach. 2025.

[16] Deepesh Data, Linqi Song, Suhas Diggavi. Data Encoding for Byzantine-Resilient Distributed Optimization. 2019.

[17] Dayue Chen, Ingram Olkin. Pao-Lu Hsu (Xu, Bao-lu): The Grandparent of Probability and Statistics in China. 2012.

[18] A. C. Davison, S. A. Padoan, M. Ribatet. Statistical Modeling of Spatial Extremes. 2012.

[19] Edsel A. Peña. Dynamic Modeling and Statistical Analysis of Event Times. 2007.

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